

A Survey on Ciphertext-Policy Attribute-Based Encryption: Performance and Security Level Analysis

Le Hoang Anh^{1,2}, Van-Hoa Nguyen^{1,2}, Thanh-Nghi Doan^{1,2},
Thanh-Nhan Huynh-Ly^{1,2}, Minh-Vi Nguyen^{1,2}, Hoai-Nam Nguyen^{1,2}

¹An Giang University, An Giang, Vietnam

²Vietnam National University, Ho Chi Minh City, Vietnam

Article Info

Article history:

Received month dd, yyyy

Revised month dd, yyyy

Accepted month dd, yyyy

Keywords:

CP-ABE

CP-ABE Survey

Access Control

Security Analysis

Performance Analysis

ABSTRACT

Ciphertext-Policy Attribute-Based Encryption (CP-ABE) has emerged as a prominent cryptographic primitive for achieving both data confidentiality and fine-grained access control in distributed environments. Despite its advantages, the practical deployment of CP-ABE remains challenging due to its high computational overhead and the limitations of existing security models, particularly in resource-constrained environments. This paper presents a comprehensive survey of CP-ABE schemes, with particular emphasis on performance optimization and security enhancement. Our analysis examines the techniques employed by existing studies to improve efficiency and strengthen security guarantees. The findings reveal that numerous approaches have been proposed to enhance the performance and security of CP-ABE; however, several open research challenges remain. This survey also discusses the suitability of different CP-ABE schemes for specific application domains. Furthermore, we categorize and compare key security-enhancing techniques, including advanced security models and pairing constructions, that have been adopted in CP-ABE schemes. Based on these findings, we identify promising future research directions for bridging the gap between theoretical CP-ABE designs and the practical requirements of modern access control systems.

This is an open access article under the [CC BY-SA](#) license.



Corresponding Author:

Le Hoang Anh

An Giang University; Vietnam National University, Ho Chi Minh City, Vietnam

Email: lhanh@agu.edu.vn

1. INTRODUCTION

The rapid growth of distributed systems, particularly in cloud computing and the Internet of Things (IoT), has created an increasing demand for secure data sharing and fine-grained access control mechanisms [1–4]. Among the various cryptographic approaches, Ciphertext-Policy Attribute-Based Encryption has emerged as a promising solution for providing data confidentiality together with flexible and fine-grained access control [5, 6]. In CP-ABE, data owners define access policies over user attributes, and only users whose attributes satisfy the specified policies can decrypt the protected data [7]. This capability enables scalable and policy-driven access control, making CP-ABE particularly suitable for distributed environments.

Leveraging these advantages, CP-ABE has attracted considerable attention in a wide range of applications, including secure cloud storage, IoT access control, electronic health record systems, and data sharing in distributed networks [3, 4, 8]. Nevertheless, the practical deployment of CP-ABE remains challenging. The computational complexity of encryption, decryption, and key generation operations often imposes substantial

overhead on resource-constrained devices. In addition, emerging application scenarios require stronger security guarantees against increasingly sophisticated adversarial models, motivating the development of more secure and efficient CP-ABE constructions.

Addressing these challenges has become a major research focus in recent years. Numerous CP-ABE schemes have been proposed to improve computational efficiency, reduce communication costs, strengthen security guarantees, and support diverse application requirements. Consequently, a systematic analysis of these developments is necessary to provide researchers and practitioners with a comprehensive understanding of the current state of the art.

Motivated by these challenges, this paper presents a comprehensive survey of CP-ABE schemes with a particular focus on performance optimization and security enhancement. We review and classify existing approaches, analyze their strengths and limitations, compare the security models and pairing constructions employed in different schemes, and discuss their suitability for various application scenarios. We further identify open research challenges and highlight promising directions for future developments in CP-ABE.

2. PRELIMINARIES

This section provides an overview of the fundamental concepts and components of Ciphertext-Policy Attribute-Based Encryption. We introduce the representation of access policies, and pairing types commonly used in CP-ABE algorithms.

CP-ABE is an advanced cryptographic scheme that not only encrypt data but also fine-grained access control. CP-ABE composed of four main algorithms such as setup, key generation, encryption, and decryption [7, 9]. The setup algorithm initializes the system by generating a public key and a master key. The key generation function produces user-specific secret keys based on their attributes. The encryption phase takes a data and an access policy to produce a ciphertext, while the decryption phase allows users to recover the plaintext if their attributes satisfy the access policy embedded in the ciphertext [7, 9]. Figure 1 illustrates the general workflow of CP-ABE, how the system components interact, and the flow of data and keys throughout the process.

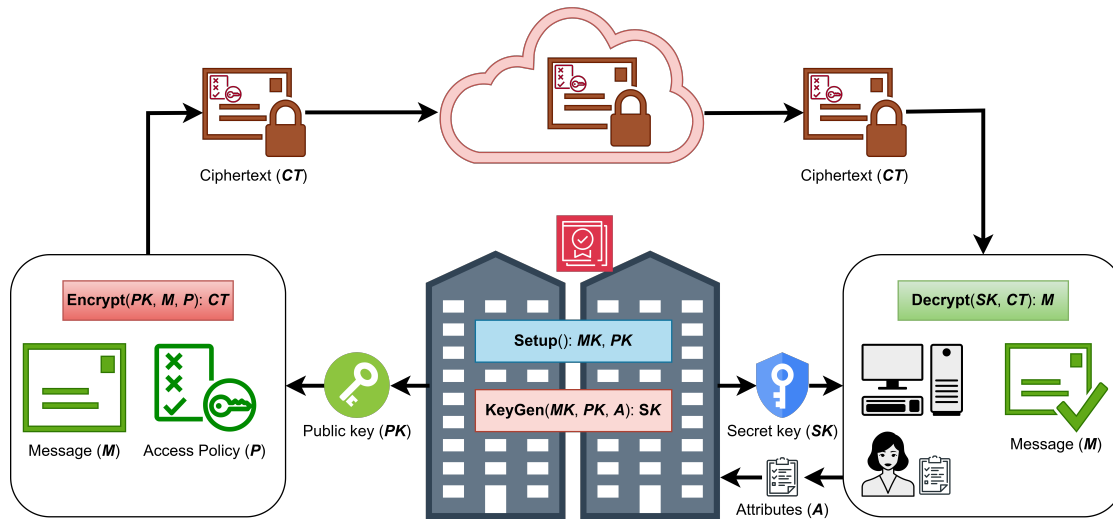


Figure 1. General workflow of CP-ABE

Data is not only protected by encryption but also controlled by access policies. Access policies can be represented using an access tree structure, where internal nodes correspond to threshold gates (AND, OR) and leaf nodes are associated with attributes [7]. This hierarchical organization allows for efficient evaluation of whether a user's attribute set satisfies the policy. Another common representation is Linear Secret Sharing Schemes (LSSS), which encode access policies in matrix form, enabling flexible and efficient enforcement of complex access structures [10].

Furthermore, the security of CP-ABE schemes is analyzed under various models, including selective and full adaptive security. Selective security requires the adversary to commit to a target access structure

before seeing the system's public parameters, while full adaptive security allows the adversary to choose the target dynamically [6, 11]. Achieving full adaptive security often involves more complex constructions and stronger assumptions, but it provides a more realistic threat model for practical applications.

Bilinear pairings are a critical component in many CP-ABE schemes, enabling the construction of advanced cryptographic protocols. Type-I pairings, also known as symmetric pairings, involve a single group and have known vulnerabilities. In contrast, Type-II and Type-III pairings are asymmetric and offer improved security and efficiency [12]. Type-III pairings are generally preferred in modern CP-ABE schemes due to their enhanced performance and security properties compared to Type-I pairings.

3. PERFORMANCE LITERATURE

Ciphertext-Policy Attribute-Based Encryption has been recognized as a vital cryptographic tool for enforcing fine-grained access control within distributed storage systems. However, traditional CP-ABE constructions often encounter efficiency bottlenecks, specifically the linear growth of ciphertext length and decryption overhead in proportion to the complexity of the access policy. Consequently, contemporary literature focuses on three primary optimization domains: constant-size ciphertext and key structures, computational costs, and access policy constructions. Table 1 provides a comparison of representative state-of-the-art CP-ABE schemes, highlighting their access structures, storage costs, and outsourcing mechanisms.

Table 1. Comparison of representative CP-ABE schemes in terms of access structures, storage costs, and outsourcing mechanisms.

Scheme	Access structure	Ciphertext size	Secret key size	Outsourcing
Yang et al. [13]	Access Tree	Constant	Linear	None
Zhang and Zhou [14]	LSSS	Linear	Linear	OD
Li et al. [15]	Access Tree	Linear	Linear	OD/OE
Guan et al. [6]	Access Tree	Constant	Linear	None
Guo et al. [11]	Access Tree	Linear	Constant	None
Malluhi et al. [5]	LSSS	Optimized	Constant	None
Agrawal and Chase [16]	LSSS	Linear	Linear	None
Luo et al. [17]	Access Tree	Linear	Linear	OD/OE
Zhang et al. [18]	Access Tree	Constant	Constant	OD
Yang et al. [10]	LSSS	Linear	Linear	OD
Sun et al. [19]	Access Tree	Linear	Linear	Online/Offline

OD: Outsourced Decryption; OE: Outsourced Encryption

- **Optimization of Storage and Transmission Overheads:** A fundamental objective in lightweight CP-ABE research is to achieve storage and transmission costs that are independent of the number of attributes involved.

Several researchers have proposed schemes that ensure a fixed ciphertext length, making them suitable for memory-limited devices [13, 18]. Yang et al. proposed a scheme, utilizing AND-gate structures can achieve a ciphertext size of approximately 1696 bits for an 80-bit security level, significantly reducing transmission bandwidth [13]. Some advancements even provide fully constant-size constructions, where both the ciphertext and the user's private key remain constant regardless of the attribute count [18].

Furthermore, some studies concentrate on reducing the key storage burden on lightweight hardware is a critical bottleneck [5, 11]. Constructions such as those proposed by Guo et al. utilize only two group elements for each secret key, enabling feasible storage on passive constraint-devices [11]. Similarly, Malluhi et al. developed decentralized schemes that maintain constant-size secret keys while supporting more flexible Disjunctive Normal Form (DNF) policies [5].

- **Optimization of Computational Costs:** To mitigate the processing strain on IoT microcontrollers, various techniques are employed to reduce the number of intensive bilinear pairing and exponentiation operations.

Outsourced computation, this strategy delegates the most compute-intensive cryptographic tasks, such as pairing operations, to semi-trusted proxy servers or fog nodes. In these models, the proxy performs a partial decryption and returns a simplified result to the user, who then needs only a single exponentiation or a minimal constant number of pairings to recover the data. Certain schemes extend this further by also outsourcing the encryption phase, thereby relieving the data sensing devices from the burden of ciphertext generation.

Another approach is online/offline processing. This approach aims to minimize real-time energy consumption, encryption is bifurcated into an offline phase, where heavy pre-computations are performed when the device is idle, and a rapid online phase that occurs once the data and access policy are identified.

- **Performance Across Specific Access Structures:** The algebraic design of the access policy directly dictates practical performance.

Schemes restricted to AND-gate policies, including those with wildcards, typically achieve constant-size results but offer lower policy expressivity [13]. While LSSS provides fine-grained and flexible control, it often leads to linear cost increases based on the number of nodes or matrix rows [6]. However, constructions like FAME (Fast Attribute-based Message Encryption) manage to fix the required pairings for decryption at a constant value regardless of the LSSS matrix complexity [16].

Recent approaches, optimized vector transformations, transform user attributes and access policies into shorter, more efficient vectors. By reducing the length of the representative vectors, these methods directly lower the number of necessary exponentiation operations during both encryption and decryption [19].

4. SECURITY LEVEL ANALYSIS

As sensitive data, ranging from industrial secrets to personal health records, is increasingly stored in semi-trusted cloud environments, the cryptographic resilience of access control mechanisms becomes paramount. This section provides a comprehensive survey of the security levels achieved by contemporary CP-ABE schemes, focusing on security models, privacy-preserving features, revocation mechanisms, and the underlying mathematical assumptions. Table 2 presents a comparison of representative CP-ABE schemes in terms of their security models, hardness assumptions, and pairing categories.

Table 2. Comparison of representative CP-ABE schemes in terms of security models, hardness assumptions, and pairing categories.

Scheme	Security Model	Hardness Assumption	Pairing Category
Yang et al. [13]	Full (Adaptive)	MSEDH	Type-III
Zhang and Zhou [14]	Selective	DLIN-based	Type-I/III
Li et al. [15]	Selective	BDH	Type-I
Guan et al. [6]	Full (Adaptive)	MSEDH	Type-III
Guo et al. [11]	Selective	aMSE-DDH	Type-I
Malluhi et al. [5]	Weak Selective	q-BDHE	Type-III
Agrawal and Chase [16]	Full (Adaptive)	DLIN	Type-III
Luo et al. [17]	Adaptive	q-BDHE	Type-I
Zhang et al. [18]	Selective	n-aMSE-DDH	Type-I
Yang et al. [10]	Full (Adaptive)	DBDH	Type-I
Sun et al. [19]	Selective	DBDH + P-DBDH	Type-III

- **Security Models:** A primary metric for evaluating the security of a CP-ABE scheme is the strength of its provable security model.

Many early lightweight constructions were proven secure only in the selective model, where the adversary is required to declare the target access structure before observing the system's public parameters [10]. For example, the schemes proposed, such as [18] and [11], utilize this model to achieve constant-size results but offer a relatively restricted security guarantee.

Recent advancements emphasize adaptive or full security, which represents a more realistic threat model where the adversary can adaptively choose the challenge access structure after querying multiple secret keys [6]. Agrawal et al. and the works of Guan et al. and Yang et al. are notable for providing full security in the standard model [6, 10, 16]. These schemes ensure that data remains confidential even if the attacker chooses their target dynamically based on the system's operational state [6, 10].

- **Indistinguishability and Active Attack Resilience:** While most CP-ABE literature focuses on indistinguishability under chosen-plaintext attacks (IND-CPA), several advanced IoT schemes target the more stringent IND-CCA (Chosen-Ciphertext Attack) security [18].

IND-CCA security ensures that the system can resist active tampering, where an adversary attempts to gain information by observing the results of decryption queries for self-constructed ciphertexts [18]. Zhang and Zhou proposed an IND-CCA-secure multi-authority scheme that integrates Zero-Knowledge Proofs (ZKP) to prevent the server from learning user secrets during authentication [14].

A fundamental security goal in CP-ABE is preventing unauthorized users from combining their different sets of attributes to decrypt a ciphertext that neither could access individually [14]. Schemes such as LiBAC and the constructions by Sun et al. and Zhang and Zhou achieve collusion resistance by binding user secrets to unique global identities (GIDs) or randomizing secret keys with unique parameters [14, 19].

- **Privacy-Preserving and Hiding Mechanisms:** In many IoT applications, the access policy itself contains sensitive information that can lead to privacy leaks if exposed to third-party storage providers [15].

To mitigate this, researchers have developed policy-hiding or hidden access structure schemes. Luo et al. proposed a scheme for the Industrial IoT (IIoT) that anonymizes attribute values using wildcards to protect commercial secrets [17]. Sun et al. introduced a fully privacy-aware approach for smart health, where both the access policy and the user's attribute set are transformed into succinct vectors, ensuring the cloud server gains zero knowledge about the underlying attributes [19].

Beyond confidentiality, security levels are enhanced through user anonymity and identity verification. Zhang and Zhou's distributed scheme utilizes Progressive Mode Attribute-Based Authentication to verify a user's legitimacy without database querying or leaking bits of the private key [14]. Guan et al. utilized Bloom filters and Chameleon hashes to prevent authorized users from maliciously leaking data, providing a layer of accountability [6].

- **Cryptographic Foundations and Assumptions:** The security level of any ABE construction is fundamentally tied to the hardness of its underlying mathematical assumptions and the pairing types used [19].

Schemes proven under standard assumptions like Decisional Linear (DLIN) or Decisional Bilinear Diffie-Hellman (DBDH) are generally considered more secure than those relying on q -type assumptions, which are more complex and less time-tested [19]. For instance, FAME is proven secure under the DLIN assumption [16], whereas the scheme by Guan et al. relies on the Multi-Sequence of Exponents Diffie-Hellman (MSEDH) assumption [6, 13].

There is a notable shift from Type-I symmetric pairings to Type-III asymmetric pairings in modern literature [14]. Type-III pairings are recommended by experts because Type-I pairings have known security vulnerabilities that prevent them from efficiently scaling to higher security levels, such as 128-bit AES [13].

5. DISCUSSION AND OPEN PROBLEMS

A clear trend is observed in the movement from linear-cost schemes toward constant-size constructions to accommodate memory-constrained IoT hardware [13]. However, a significant trade-off persists between efficiency and flexibility: constant-size schemes are often limited to AND-gate structures, whereas more expressive LSSS or tree-based models generally result in larger ciphertexts [15].

The mechanism of outsourced computation provides a definitive solution for mobile IoT devices, allowing them to perform decryption in constant time regardless of the back-end complexity [18]. Furthermore, the integration of Type-III asymmetric pairings with online/offline computation is currently the gold standard

for developing high-performance, adaptively secure systems for sectors like industrial IoT or smart healthcare, where both low latency and robust security are mandatory [13].

The evolution of CP-ABE for the IoT indicates a clear trajectory toward adaptive security and active attack resilience (IND-CCA) [6]. While earlier schemes prioritized performance by adopting selective security models and symmetric pairings, modern requirements for high-stakes environments like e-health and IIoT have necessitated the use of Type-III asymmetric pairings and policy-hiding mechanisms [6].

A persistent challenge remains the escrow problem inherent in centralized systems; however, the emergence of decentralized, multi-authority schemes that avoid a single point of failure provides a robust alternative for distributed IoT networks [14]. Future research should continue to focus on achieving these high security levels without introducing linear computational growth, ensuring that even the most constrained sensors can participate in secure, fine-grained data ecosystems [13].

6. CONCLUSION

This paper has presented a comprehensive survey of CP-ABE schemes, highlighting the techniques adopted in existing studies and discussing the suitability of different schemes for specific application domains. Our analysis reveals that significant performance improvements have been achieved through approaches such as outsourcing computationally intensive operations to external entities and optimizing access policy representations. Furthermore, numerous techniques have been proposed to enhance security, including the adoption of fully adaptive security models, resistance against active attacks, and the use of alternative pairing constructions. Despite these advances, several challenges remain in balancing efficiency, security, and practicality in resource-constrained environments. The findings and research directions presented in this survey provide a valuable foundation for future research on secure and efficient CP-ABE schemes.

ACKNOWLEDGMENTS (IF APPLICABLE)

The authors would like to thank An Giang University and Vietnam National University, Ho Chi Minh City (VNU-HCM) for their financial support and research facilities.

FUNDING INFORMATION (MANDATORY)

This research is funded by Vietnam National University HoChiMinh City (VNU-HCM) under grant number C2026-16-24.

AUTHOR CONTRIBUTIONS STATEMENT (MANDATORY)

This journal uses the Contributor Roles Taxonomy (CRediT) to recognize individual author contributions, reduce authorship disputes, and facilitate collaboration.

Name of Author	C	M	So	Va	Fo	I	R	D	O	E	Vi	Su	P	Fu
Le Hoang Anh	✓	✓	✓	✓	✓	✓		✓	✓	✓			✓	
Van-Hoa Nguyen	✓	✓		✓	✓	✓		✓	✓	✓		✓		
Thanh-Nghi Doan	✓	✓		✓	✓	✓			✓	✓		✓		
Thanh-Nhan Huynh-Ly						✓			✓	✓				
Minh-Vi Nguyen				✓					✓	✓				
Hoai-Nam Nguyen			✓						✓	✓	✓			

C : Conceptualization

M : Methodology

So : Software

Va : Validation

Fo : Formal Analysis

I : Investigation

R : Resources

D : Data Curation

O : Writing - Original Draft

E : Writing - Review & Editing

Vi : Visualization

Su : Supervision

P : Project Administration

Fu : Funding Acquisition

CONFLICT OF INTEREST STATEMENT (MANDATORY)

Authors state no conflict of interest.

REFERENCES

- [1] J. Domingo-Ferrer, O. Farràs, J. Ribes-González, and D. Sánchez, "Privacy-preserving cloud computing on sensitive data: A survey of methods, products and challenges," *Computer Communications*, vol. 140-141, pp. 38–60, 2019. [Online]. Available: <https://www.sciencedirect.com/science/article/pii/S0140366418310740>
- [2] S. Swain, P. K. Pattnaik, and B. Dash, "A review on privacy preservation in cloud computing and recent trends," in *Micro-Electronics and Telecommunication Engineering*, D. K. Sharma, S.-L. Peng, R. Sharma, and G. Jeon, Eds. Singapore: Springer Nature Singapore, 2024, pp. 365–376.
- [3] H. M. A. Ahmed, S. Zubair, and M. Tawfik, "A privacy-preserving cloud storage framework with hybrid encryption, homomorphic keyword search, and blockchain-based integrity verification," *Scientific Reports*, vol. 16, p. 16697, 4 2026.
- [4] D. Ziegler, J. Sabongui, and G. Palfinger, *Fine-Grained Access Control in Industrial Internet of Things*. Springer Science+Business Media, 01 2019, pp. 91–104. [Online]. Available: https://doi.org/10.1007/978-3-030-22312-0_7
- [5] Q. Malluhi, A. Shikfa, V. Tran, and V. Trinh, "Decentralized ciphertext-policy attribute-based encryption schemes for lightweight devices," *Computer Communications*, vol. 145, pp. 113–125, 9 2019.
- [6] Z. Guan, W. Yang, L. Zhu, L. Wu, and R. Wang, "Achieving adaptively secure data access control with privacy protection for lightweight IoT devices," *Science China Information Sciences*, vol. 64, p. 162301, 6 2021.
- [7] L. H. Anh, Y. Kawada, S. Huda, M. A. Ali, Y. Koderu, and Y. Nogami, "An implementation of ELiPS-based Ciphertext-Policy Attribute-Based Encryption," in *2023 Eleventh International Symposium on Computing and Networking Workshops (CANDARW)*. IEEE, 11 2023, pp. 220–226.
- [8] E. Greene, P. Proctor, and D. Kotz, "Secure sharing of mhealth data streams through cryptographically-enforced access control," *Smart Health*, vol. 12, pp. 49–65, 04 2018. [Online]. Available: <https://doi.org/10.1016/j.smhl.2018.01.003>
- [9] J. Shi, Q. Yu, Y. Yu, L. Wang, and W. Zhang, "Privacy protection in social applications: A ciphertext policy attribute-based encryption with keyword search," *International Journal of Intelligent Systems*, vol. 37, pp. 12 152–12 168, 09 2022. [Online]. Available: <https://doi.org/10.1002/int.23080>
- [10] Y. Yang, X. Liu, and R. H. Deng, "Lightweight Break-Glass Access Control System for Healthcare Internet-of-Things," *IEEE Transactions on Industrial Informatics*, vol. 14, pp. 3610–3617, 8 2018.
- [11] F. Guo, Y. Mu, W. Susilo, D. S. Wong, and V. Varadharajan, "CP-ABE With Constant-Size Keys for Lightweight Devices," *IEEE Transactions on Information Forensics and Security*, vol. 9, pp. 763–771, 5 2014.
- [12] L. H. Anh, Y. Kawada, S. Huda, Y. Koderu, and Y. Nogami, *Performance Analysis of ELiPS-Based CP-ABE with Optimized Decryption Functions*. Springer Nature, 01 2024, pp. 345–354. [Online]. Available: https://doi.org/10.1007/978-981-97-5810-4_30
- [13] W. Yang, R. Wang, Z. Guan, L. Wu, X. Du, and M. Guizani, "A Lightweight Attribute Based Encryption Scheme with Constant Size Ciphertext for Internet of Things," in *ICC 2020 - 2020 IEEE International Conference on Communications (ICC)*. IEEE, 6 2020, pp. 1–6.
- [14] Z. Zhang and S. Zhou, "A decentralized strongly secure attribute-based encryption and authentication scheme for distributed Internet of Mobile Things," *Computer Networks*, vol. 201, p. 108553, 12 2021.

-
- [15] R. Li, C. Shen, H. He, X. Gu, Z. Xu, and C.-Z. Xu, “A Lightweight Secure Data Sharing Scheme for Mobile Cloud Computing,” *IEEE Transactions on Cloud Computing*, vol. 6, pp. 344–357, 4 2018.
 - [16] S. Agrawal and M. Chase, “FAME: Fast Attribute-based Message Encryption,” in *Proceedings of the 2017 ACM SIGSAC Conference on Computer and Communications Security*. ACM, 10 2017, pp. 665–682.
 - [17] W. Luo, Z. Lv, L. Yang, G. Han, and X. Zhang, “FOC-PH-CP-ABE: An Efficient CP-ABE Scheme With Fully Outsourced Computation and Policy Hidden in the Industrial Internet of Things,” *IEEE Sensors Journal*, vol. 24, pp. 28 971–28 981, 9 2024.
 - [18] Z. Zhang, W. Zhang, and Z. Qin, “Fully Constant-Size CP-ABE with Privacy-Preserving Outsourced Decryption for Lightweight Devices in Cloud-Assisted IoT,” *Security and Communication Networks*, vol. 2021, pp. 1–16, 5 2021.
 - [19] J. Sun, H. Xiong, X. Liu, Y. Zhang, X. Nie, and R. H. Deng, “Lightweight and Privacy-Aware Fine-Grained Access Control for IoT-Oriented Smart Health,” *IEEE Internet of Things Journal*, vol. 7, pp. 6566–6575, 7 2020.